UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

End of Term 1 Exam Coverage	هيكل اختبار نهاية الفصل الدراسي الأول
Grade 11 Advanced - Chemistry	**الصف الحادي عشر - متقدم - الكيمياء**
Academic Year 2026-2027	العام الدراسي 2026-2027
Inspire	انسباير

عدد الأسئلة Number of questions	درجة السؤال Marks per question	الدرجة القصوى الممكنة Maximum overall grade
30	**(2,3,4)**	**100**

مدة الاختبار Exam duration	طريقة التطبيق Mode of implementation	الآلة الحاسبة Calculator
120 دقيقة **120 minutes**	**إلكتروني** **Electronic**	**يُسمح باستخدام الآلة الحاسبة العلمية** **The use of a scientific calculator is allowed**

⚠ Questions might appear in a different order in the actual exam.

Q	Learning outcome	Reveal (English)	Page
Q1	CHM.5.5.01.001.07 Perform interconversion between units of temperature and heat	Textbook + Table 1+ Example problem 1+ Practice Problems	5, 6
Q2	CHM.5.5.01.001.08 Describe the relationship between the specific heat capacity of a substance and the resistance to change in temperature	Textbook + Table 2+ Practice Problems	7, 8, 9
Q3	CHM.5.5.01.002.01 Calculate the amount of heat absorbed by a substance as its temperature changes	Textbook + Example problem 2+ Practice Problems	7, 8, 9
Q4	CHM.5.5.01.002.02 Calculate the amount of heat released by a substance as its temperature changes	Textbook + Example problem 2+ practice problems	7, 8, 9
Q5	CHM.5.5.01.002.03 Calculate the specific heat capacity of a sample given its mass and temperature change	Textbook + Table 2 + Example problem 2+ Practice Problems	8, 9
Q6	CHM.5.5.01.004.01 Define thermochemistry	Textbook	14

Q	Learning outcome	Reveal (English)	Page
Q7	CHM.5.5.01.004.02 Differentiate among the system, the surroundings, and the universe	Textbook + Figure 7	14
Q8	CHM.5.5.01.006.03 Perform and conduct an experiment to determine the enthalpy of a chemical reaction	Textbook + Figure 7	14
Q9	CHM.5.5.01.004.04 Compare and contrast potential energy diagrams of exothermic and endothermic reactions in terms of general shape, enthalpy of reactants and products, activation energy of forward and backward reactions, and enthalpy of reaction and its sign	Textbook + Figures 8, 9	15, 16, 17
Q10	CHM.5.5.01.006.02 Define enthalpy change of reaction (ΔHrxn) while writing the equation used	Textbook + Figures 8, 9	15, 16, 17
Q11	CHM.5.5.01.006.09 Perform calculations using the enthalpy of combustion	Textbook + Example problem 4 + Practice Problems	19, 21
Q12	CHM.5.5.01.006.08 Write a thermochemical equation for the combustion of a compound, while including the sign of ΔH and identifying whether the reaction is exothermic or endothermic	Textbook + Table 3	18, 19
Q13	CHM.5.5.01.003.01 Define molar enthalpy (heat) of vaporization, ΔHvap CHM.5.5.01.003.03 Define molar enthalpy (heat) of condensation, ΔHcond	Textbook+ Table 4 + Figure 10	19, 20
Q14	CHM.5.5.01.003.02 Define molar enthalpy (heat) of fusion, ΔHfus CHM.5.5.01.003.04 Define molar enthalpy (heat) of solidification, ΔHsolid	Textbook+ Table 4 + Figure 10	19, 20
Q15	CHM.5.5.01.003.05 Perform phase-change calculations using enthalpy of fusion, vaporization, condensation, and solidification	Textbook + Figure 10 + Practice problems	20, 21
Q16	CHM.5.5.02.002.01 Calculate, using Hess's law, the ΔH of a reaction	Textbook + Figure 13 + Example problem 5 + Practice Problems	23, 24, 25, 26
Q17	CHM.5.5.02.006.01 Define the standard enthalpy of formation of a compound, ΔHfo, while determining on what basis it is defined	Textbook	26, 27, 28
Q18	CHM.5.5.02.006.02 Identify the standard enthalpy (heat) of formation of a free element in its standard state	Textbook+ Figure 15	28
Q19	CHM.5.5.02.006.03 Calculate, using the standard enthalpies of formation, the enthalpy of a reaction, ΔHrxn	Textbook + Example problem 6 + Practice Problems	28, 29, 30

Q	Learning outcome	Reveal (English)	Page
Q20	CHM.5.5.02.006.04 Calculate the enthalpy of reaction (ΔHrxn) using standard enthalpies of formation of products and reactants while using the sign of ΔH for determining if the reaction is exothermic or endothermic	Textbook + Example problem 6 + Practice Problems	29, 30
Q21	CHM.5.4.01.020.01 Calculate the average reaction rate using the rate of consumption of reactants or the rate of production of products	Textbook + Example problem 1 + Practice Problems	46, 47, 48, 49
Q22	CHM.5.4.01.002.02 Explain why a collision between two particles is necessary for a reaction to occur	Textbook + Figures 3 and 4 + Table 1	49, 50, 51
Q23	CHM.5.4.01.022.01 Define activated complex	Textbook + Figures 5 and 6	51, 52
Q24	CHM.5.4.01.022.03 Describe the relationship between activation energy and the rate of the reaction	Textbook + Figures 5 and 6 + Table 1	51, 52
Q25	CHM.5.4.01.022.04 Interpret exothermic and endothermic reactions using the collision theory	Textbook+ Figures 5, 6	51, 52
Q26	CHM.5.4.01.005.09 Write the rate law for a chemical reaction at a given temperature	Textbook + Practice Problems	62, 63, 64, 65
Q27	CHM.5.4.01.006.02 Use the method of initial rates to write the rate law of the reaction	Textbook + Table 2 + Practice Problems	64, 65, 66
Q28	CHM.5.4.01.006.03 Use the method of initial rates to calculate the specific rate constant and identify its unit	Textbook	63, 64
Q29	CHM.5.4.01.006.04 Use the method of initial rates to calculate the concentration of reactants	Textbook + Table 2 + Practice Problems	63, 64, 65, 66
Q30	CHM.5.4.01.006.05 Use the method of initial rates to calculate the initial rate of the reaction	Textbook + Table 2 + Practice Problems	63, 64, 65, 66